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Software Requirements Specification (SRS)

Smart Attendance System Using QR Code

1. Introduction

1.1 Purpose

The purpose of this document is to specify the functional and non-functional requirements of the Smart Attendance System Using QR Code. The system automates attendance management by allowing teachers to generate QR codes and students to scan them to mark attendance.

This SRS is intended for students, teachers, administrators, developers, testers, and project supervisors.

1.2 Scope

The Smart Attendance System enables students to mark attendance through QR code scanning. Teachers can generate class-specific QR codes, monitor attendance records, and generate reports. Administrators can manage users, courses, and system activities.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
QR	Quick Response Code
UI	User Interface
DB	Database
OTP	One-Time Password
API	Application Programming Interface

1.4 References

- IEEE 830 / IEEE 29148 Standard
- Software Engineering – Ian Sommerville
- QR Code Technology Documentation

1.5 Overview

This document describes the overall system, functional requirements, interface requirements, and non-functional requirements of the Smart Attendance System.

2. Overall Description

2.1 Product Perspective

The system is a web-based or mobile-based attendance management application that uses QR code technology to record student attendance automatically.

2.2 Product Functions

- User registration and authentication
- QR code generation
- QR code scanning

- Attendance recording
- Attendance report generation
- Course and class management
- Attendance history tracking
- Admin dashboard and monitoring

2.3 User Classes and Characteristics

User	Description
Student	Scans QR codes and views attendance records.
Teacher	Creates classes, generates QR codes, and manages attendance.
Admin	Manages users, courses, and system reports.

2.4 Operating Environment

- Web Browser (Chrome, Firefox, Edge, Safari)
- Android Mobile Device
- Server OS: Windows/Linux
- Database: MySQL

2.5 Design and Implementation Constraints

- Secure login required
- Internet connection required
- QR codes must be time-limited
- Must support multiple users simultaneously

2.6 User Documentation

- Student User Manual
- Teacher User Manual
- Administrator Manual

2.7 Assumptions and Dependencies

- Users have internet access
- Students possess smartphones with cameras
- Database server is available

3. System Features (Functional Requirements)

3.1 User Authentication

Description:

The system shall authenticate users according to their roles.

Functional Requirements:

- FR-1: The system shall allow users to log in using email and password.
- FR-2: The system shall identify Student, Teacher, and Admin roles.

3.2 QR Code Generation

Description:

Teachers generate unique QR codes for attendance.

Functional Requirements:

- FR-3: Teachers shall generate QR codes for each class session.
- FR-4: QR codes shall expire after a specified time.

3.3 QR Code Attendance

Description:

Students scan QR codes to record attendance.

Functional Requirements:

- FR-5: Students shall scan QR codes using mobile devices.
- FR-6: The system shall verify and record attendance automatically.

3.4 Attendance Management

Description:

Teachers manage attendance records.

Functional Requirements:

- FR-7: Teachers shall view attendance records.
- FR-8: Teachers shall update attendance when necessary.

3.5 Reports and Analytics**Description:**

The system generates attendance reports.

Functional Requirements:

- FR-9: Generate daily attendance reports.
- FR-10: Generate student attendance summaries.

3.6 Administration**Description:**

Administrators manage the overall system.

Functional Requirements:

- FR-11: Manage users and courses.
- FR-12: Generate system activity reports.

4. External Interface Requirements

4.1 User Interface

- Web-based graphical interface
- Mobile-friendly design
- Simple and user-friendly layout

4.2 Hardware Interfaces

- Server hardware
- Smartphone with camera

- PC/Laptop/Tablet

4.3 Software Interfaces

- MySQL
- QR Code Library
- Node.js / PHP / Python Backend
- Web Browser

4.4 Communication Interfaces

- HTTP/HTTPS Protocol
- Internet Connectivity

5. Non-Functional Requirements

5.1 Performance Requirements

- The system shall support at least 500 concurrent users.
- Attendance processing time shall be less than 3 seconds.

5.2 Security Requirements

- Secure login and password encryption.
- Role-based access control.
- Protection of attendance records.

5.3 Reliability Requirements

- System availability shall be 99%.
- Attendance records shall be stored accurately.

5.4 Usability Requirements

- Easy to learn and use.
- Clear navigation and error messages.

5.5 Portability Requirements

- Accessible from different browsers and devices.

6. Other Requirements

- QR codes shall be unique for each class session.
- Attendance data must be protected.
- Students cannot mark attendance after QR code expiration.

7. Appendices

7.1 Appendix A: Data Flow Diagram (Level 1)

Student → [Login] → Authentication

Teacher → [Generate QR Code] → QR Service

Student → [Scan QR Code] → Attendance Verification

Attendance Verification → Attendance Database

Teacher → Attendance Report

Admin → System Monitoring

7.2 Appendix B: Entity Relationship Diagram (ERD)

USER

- user_id (PK)
- name
- email
- password
- role

COURSE

- course_id (PK)
- course_name
- teacher_id (FK)

QR_CODE

- qr_id (PK)
- course_id (FK)
- generated_time
- expiry_time

ATTENDANCE

- attendance_id (PK)
- user_id (FK)
- course_id (FK)
- attendance_date
- status